

# mBERT vs MuRIL for Hinglish / Hindi-English Harmful Speech Detection

One-page research summary for review and project defense

## Research Question

Does Indian-language-specific pretraining in MuRIL improve hate/offensive speech detection for Hinglish and Hindi-English code-mixed text compared with general multilingual pretraining in mBERT? Current evidence says the answer is conditional: the better model changes with dataset situation.

## Datasets

| Dataset              | Rows   | Positive label | Situation                                  |
|----------------------|--------|----------------|--------------------------------------------|
| Kaggle Hinglish Hate | 4,780  | hate           | controlled Hinglish subset from Kaggle     |
| CM Code-mixed        | 3,900  | offensive      | Indian politics / Twitter/X, hate-adjacent |
| THAR Religion        | 11,549 | AntiReligion   | YouTube targeted religious hate            |

## Methods

- Models: mBERT and MuRIL fine-tuned as binary classifiers.
- Matched setting: train and evaluate on the same dataset situation.
- Robustness setting: evaluate trained checkpoints across other datasets and mixed-training conditions.
- Primary metric: Macro F1; positive recall is tracked because false negatives are missed harmful examples.

## Main Matched Multi-Seed Result

| Dataset | Better model | mBERT Macro F1 | MuRIL Macro F1 | Meaning                                      |
|---------|--------------|----------------|----------------|----------------------------------------------|
| Kaggle  | mBERT        | 67.5 +/- 2.1   | 58.1 +/- 5.7   | mBERT clear win; MuRIL misses more positives |
| CM      | mBERT narrow | 77.7 +/- 1.9   | 76.1 +/- 2.3   | both competitive; labels are offense-like    |
| THAR    | MuRIL        | 74.7 +/- 0.1   | 76.5 +/- 1.3   | MuRIL wins targeted religious hate           |

## Current Interpretation

The central evidence is that model ranking changes by dataset: mBERT wins Kaggle, mBERT narrowly wins CM, and MuRIL wins THAR. Cross-dataset robustness is weak, so hate, offensive, and AntiReligion labels must not be treated as interchangeable. The 79-row benchmark remains excluded from primary claims because its provenance is uncertain.

## Limitations

- Mixed-training and cross-dataset transformer results are mostly single-seed evidence.
- Label definitions differ across datasets, which limits universal model-superiority claims.
- Dataset licenses/citation details need final verification before public final release.
- More manual error examples are needed to explain failure modes in the final draft.

## Feedback Requested

Is the conditional claim clear enough? Are the dataset caveats acceptable? Which result should be prioritized for the next multi-seed run: mixed all-three training, Kaggle+THAR, CM+THAR, or cross-dataset transfer?